

Onboarding



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User creation

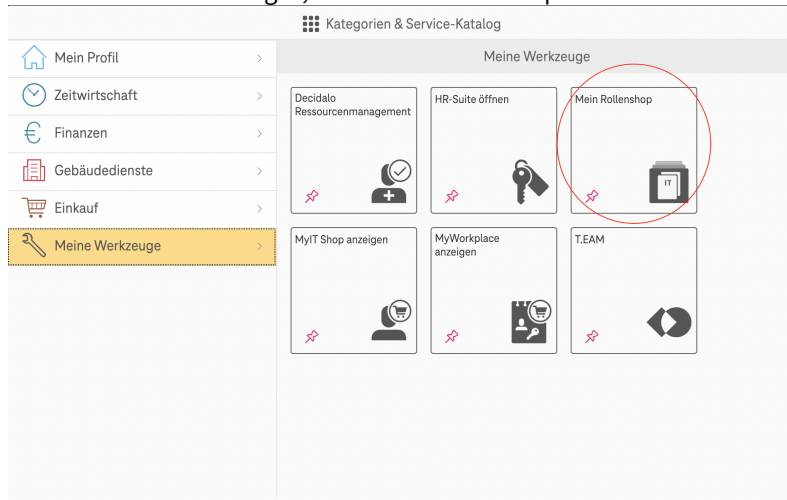
User creation for users with telekom.de, telekom.com and t-systems.com eMail address:

Important Note: In case you already have a self-created (unmanaged) Google Account and have not yet accepted the invitation to manage this account, you should do so first, otherwise this account will be renamed automatically. You have such an account and have not received an invitation? Please contact the [GCLZ Support](#).

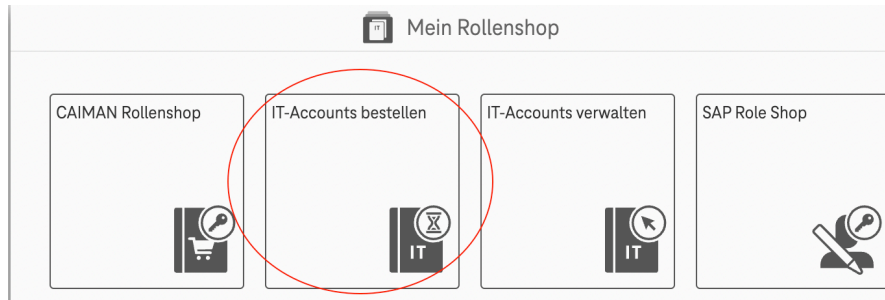
To get a user in Google Cloud you need to order the Role via MyPortal Roleshop:

1. Go to MyPortal: <https://myportal.corp.telekom.de/>

2. Select: "Meine Werkzeuge", and "Mein Rollenshop"



3. Select "IT-Accounts bestellen"



4. Select who should receive the access



5. Search for 'GCP@DTIT 'and the bottom left corner, and choose GCLZreplication on the right side

Ihre Rollenbestellung

1. Mitarbeiter: [Ändern](#)

2. Welche Rolle(n) möchten Sie bestellen?

Sie haben noch keine Rollen hinzugefügt.

Bitte geben Sie den Namen der Anwendung, den Rollennamen, die Rollen-ID oder Suchbegriffe ein, um Ihre Rolle zu finden.

Dank Ihrer Eingabe wurden 2 Ergebnisse gefunden. Bitte treffen Sie Ihre Auswahl durch Anklicken oder grenzen Sie die Suche stärker ein.

Rollename	Anwendung
 GCLZreplication	GCP@DTIT
 GCLZreplicationAG	GCP@DTIT

[Weiter zum nächsten Schritt](#)

Kennen Sie die Anwendung, für die Sie die Rolle(n) benötigen?

Anwendung

6. Fill the form and finish the order process

Hier können Sie bei Bedarf weitere Angaben zur Bestellung ergänzen

Bitte teilen Sie uns mit, warum Sie die Rolle(n) benötigen *

▼ GCLZreplication GCP@DTIT

Möchten Sie die Rolle zeitlich begrenzen?

Hinweis: Wird kein Gültigkeitszeitraum ausgewählt, wird die Rolle zeitlich unbeschränkt bestellt.

Gültig von bis

* Pflichtfeld

[Weiter zum nächsten Schritt](#)

Note: It can take up to an hour to log in to Google with the account.

Video:

https://yamtube.telekom.de/media/01%20MyPortal%20Role/0_he984ljj

User creation for Users with other eMail address (e.g. t-mobile.pl)

At the moment user onboarding is a manual process. Soon it will work over an self service portal, we will announce here.

For now please get in contact with gclz-request@telekom.de

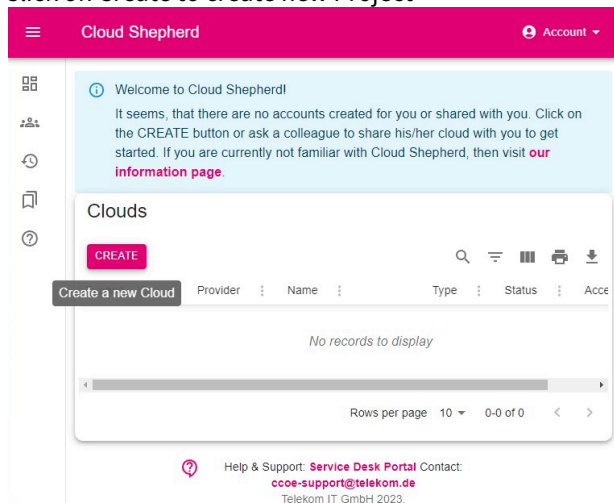
Project creation

Project creation for users with telekom.de, telekom.com and t-systems.com eMail address:

Prerequisite: You must have a GCP user as described in the "User creation" section.

For project creation you can use Cloud Shepherd as a Self Service Portal.

1. Login to the Cloud Shepherd Portal: <https://cs.telekom.de/>
2. Click on Create to create new Project



3. Select **Cloud Provider** as GCP, the **Project Type** (Development or Production), and Fill the **Environment Name**, Prefix (optional), **Cost Object**

Cloud Shepherd Account

Advanced settings

Members

As cloud owner you will be member of the newly created project by default.

Cost Object *

Labels

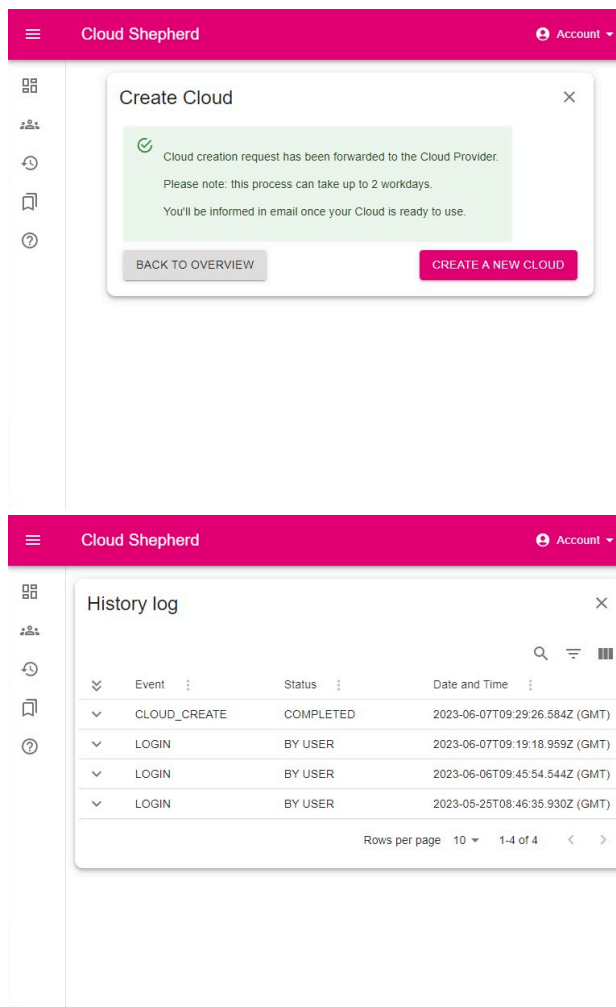
Key

Value

CANCEL SUBMIT

Cost Object is a PSP element in following format: **T-XXX-XXXXXXXXXX-XX-XXXX**

4. Optionally, you can add **Members** who will automatically get access to the project as well. Keep in mind, that all users need to have a GCP user, like described in the "User creation" section.
5. You can add further optional [optional values](#), then click on **"Submit"**
6. Once you submitted your request the Portal will send the data to the Provider. Wait until you get a green message, that the request has been successfully forwarded. You will see the project creation request in the History log as well.



- Once the Project created you will get an information in eMail from FMB GCLZ-Request with your project ID and URL

Project creation for Users with other eMail address (e.g. t-mobile.pl)

At the moment project onboarding is a manual process. Soon it will work over an self service portal, we will announce here.

For now please get in contact with gclz-request@telekom.de

Video of the manual:

https://yamtube.telekom.de/playlist/dedicated/0_eiel7x63/

Privacy Information

General

DEUTSCHE TELEKOM SECURITY GMBH attaches great importance to protecting your personal data. It is important to us to inform you what personal data we collect, how your data is processed

1. Where can I find the information that is important to me?

More information on data privacy, in particular on intended use, deletion periods, etc. can be found in these complementary data privacy information. You can send a request for information in accordance with Art. 15 GDPR for this specific data processing to (gclz-request@telekom.de). The general information on your rights, the data privacy officer, etc. is available in the data privacy notes for employees at [myDMS-Document](#).

2. Who is responsible for data processing?

DEUTSCHE TELEKOM SECURITY GMBH Bonnertalweg 100, 53113 Bonn acts as the data controller.

3. What data is recorded, how is it used, and how long is it stored?

E-mail address for the user identification

Name and IP address for identification

All data Google cloud Landingzone team needs for operation are deleted latest after 90 days.

4. Who does Deutsche Telekom pass my data on to?

Google Cloud Identity to enable users to login.

Owing to legal obligations: In certain cases, we are legally obliged to transfer data to a state authority that requests it.

5. Where is my data processed?

Your data will be processed in Europe.

More about the data privacy information by using google services is to find by google directly <https://policies.google.com/privacy?hl=de>

Operational Level Agreement (OLA)



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1. OLA Foundation



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1 Introduction

With the Google Cloud Platform (GCP) Landing Zone, the ‘Cloud center of excellence’ (CCOE) within Deutsche Telekom IT provides a secure Infrastructure as Code (IaC) environment on the Google Cloud Platform (GCP), for the operation of DTIT applications and workloads as a service. It supports compliance with the corporate requirements, regarding data privacy and security and follow GCP best practices. Among other use cases, the solution focuses on Google Analytics Solutions for BI and data driven businesses with open-source-affinity. The use of the Landing Zone is highly automated, especially for Account and Project Creation by using a Self-Service Portal & automated cost distribution into DTAG.



Mission/Vision

Our Motivation to establish Google as an alternative Hyperscaler at Deutsche Telekom AG for mainly Cloud Native, Data Analytics BI & AI Use Cases is, that we strongly believe in enabling business by adding this value throughout the entire group.

2 Scope of Essentials Service

Within the DTIT CCoE GCP Landing Zone, Google is responsible for providing basic protection of the infrastructure offered and running through Google Cloud Platform (GCP). This infrastructure consists of the hardware, software, network, and facilities on which and in which Google Cloud Services run.

The Landing Zone provides functionalities on several levels:

Provisioning of GCP accounts/environments for projects/applications according to their needs (e.g. different environments like development or production, connectivity requirements etc.),

[Identity provisioning, most importantly federation with the DTAG Active Directory to authorize access to GCP accounts.](#)

[Central logging and monitoring of LZ management \(CI/CD, provisioning, billing, security management, etc..\) with recording of, among others, all GCP API activities on LZ Management Level in the landing zone and providing means to protect and analyze that data while ensuring privacy and security requirements.](#)

Note that the DTIT CCoE GCP Landing Zone does not contain servers (neither virtual machines nor bare metal). Rather it consists of resources such as accounts, VPCs, IAM roles, Compute Engine, Cloud Firewall, Cloud Load Balancing, Pub/Sub topics or Cloud Storage buckets, to name just a few, leveraging the respective fully-managed services from GCP. By using this environment, the DTAG applications can go-live on GCP faster. They are running in an approved environment compliant with all requirements and, most importantly, application teams can focus on delivering business value, as they are being freed from heavy lifting regarding connectivity, billing or implementing the required controls.

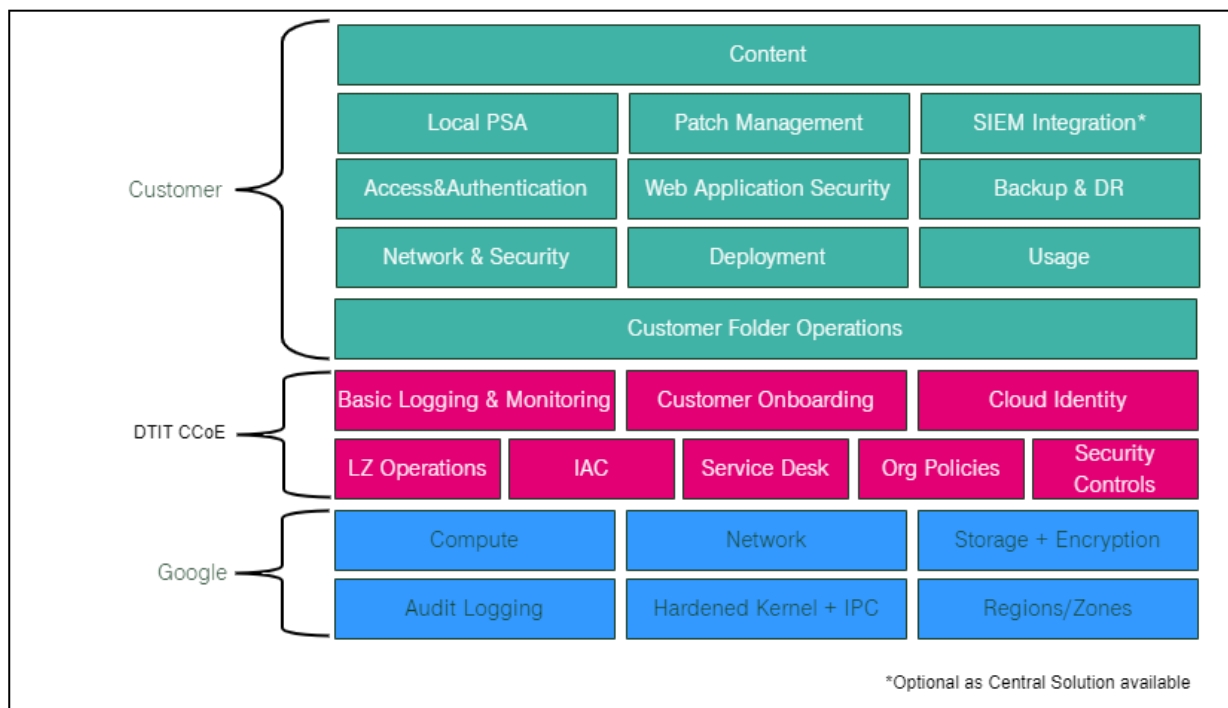
[Scope of the document:](#)

The “GCP Landing Zone Foundation” offers a professional managed service that consist of well-architected Blueprints and Core Services provided by DTIT CCoE. That allows customers to basically create projects, infrastructure and environments that are configured in line with security policies, compliance guidelines and cloud native best practices. Scope of this document is to describe the "Foundation Services" with appropriate level of details.

Based on the mandatory Foundation Service, DTIT CCoE delivers a rich set of optional services that are not in scope of this document.

2.1 Shared Responsibility Model (Foundation)

The shared responsibility model regulates the responsibilities and accesses of the different parties within the Google Cloud Landing Zone of the DTIT.



Google

Google is responsible for the security of the infrastructure on which all services offered by Google Cloud are running. This infrastructure consists out of hardware, software, network as well as the Datacenter on and in which the Google Cloud services are running.

DTIT CCoE

With the Google Cloud Foundation Service DTIT offers the basis for rapid affiliate onboarding to support other entities' projects to the greatest extent possible. In detail, it includes the following building blocks:

- **Customer Onboarding** - Our GCP resource hierarchy is designed to serve multiple Legal Entities. Based on that standard, individual customer parameter need to be identified, aligned and configured just for the Customer Folder (e.g. Network Parameter, Production Level) as also an Affiliate Agreement has to be signed and a Billing Account created.
- **Landing Zone Operations** - The landing zone operation includes the constant maintenance and care of the landing zone. Both at the code base and in particular at the continuous adjustment of the security parameters to the current threat situation.
- **Cloud Identity** - Cloud Identity is used to manage Google Cloud Platform users from a central location in the Google Admin console (admin.google.com). DTIT CCoE is responsible for the maintenance and care of this system as well as the connection of the local identity providers.
- **Basic Logging & Monitoring** - The Google Landing Zone is monitored at the top level to detect security incidents at a central location to be able to react quickly. The customers are responsible for the monitoring, logging and security management of their own customer folders, as they have the full control of their folders.

Customer

Customers are asked to regularly check their contractual obligations to cooperate (see 3.) in order to obtain a comprehensive overview. Irrespective of this, the following list should provide important information for the daily work with the Google Cloud Landing Zone:

- Customer Folder Operations - After the Customer Onboarding where DTIT CCoE enables the individual Landing Zone for the customer, you are responsible for the operation and all future changes to this local environment.
- Access & Authentication - Please have in mind that you are responsible for the management of those users who are granted access to the self-service portals and your virtual machines. Please review carefully and regularly who is granted access to. Make use of strict password policies and check regularly the access control lists.
- Network & Security - Please consider that you are responsible for the management of your Network Tiering & Topology including a CIDR Pool Allocation Management of your customer folder.
- Backups & DR - By default your data and cloud resources will not be backed up and cannot be recovered within the self-service portal. Please ensure that you possess suitable tools and back up your data by yourself.
- Patch Management - Please ensure that your operating systems as well as applications are always up to date. In addition, make use of the solution's flexibility to define when your organization can tolerate potential patch-related outages best.
- Usage - In general, customers are responsible for the capacity management and with this the cost of their Environment. Cost management is the key to successful cloud infrastructure operation.
- Order data processing – Please survey carefully if you process personal data on your GCP Environment.

2.2 Landing Zone Operations by DTIT CCoE

2.2.1 Service Desk

DTIT CCoE Support services are provided within the European Union. The GCP services itself are available worldwide according to the customer's choice.

DTIT CCoE Support is the central contact partner for the customer's key users for incidents and general service requests. Support is available to the customer workdays 9h-16h. and can be contacted as follows:

- A ticket can be raised on <https://jira.telekom.de/servicedesk/customer/portal/301/group/2898>
- alternatively via e-mail GCLZ-Support@telekom.de -> connected to our Ticket system for Service Request

2.2.2 Operating times

- Operating times (Uptime) refers to the period during which the GCP service is available. Uptime is 7 x 24 hours.
- DTIT CCoE Attended Operation Time (AOT) is the period during which technical support is available and during which incidents affecting system availability is handled. The AOT period is equivalent to the OFFICE TIME. The OFFICE TIME is Mondays to Fridays, excluding national holidays (Germany), from 09:00 to 16:00
- Response time 8 h within the service time zone

2.2.3 Google Support

The DTIT CCoE Cloud Master Agreement (CMA) ensures [Google Premium Support Level](#). Premium Support is a paid support offering designed for enterprises that run mission critical workloads and require fast response times, platform stability, and increased operational efficiencies. Within the Affiliate Agreement, customer take advantage by joining this support level. Premium Support will be invoiced on a consumption based, shared cost model. If the customer is in a permission group

higher than Viewer in a project or folder, he has the possibility to open a support case in the Google Console for his project.

- As a Premium Support customer, you are assigned to a shared, named DTAG Technical Account Manager (TAM). Technical Account Managers are trusted technical advisors that focus on operational rigor, platform health, and architectural stability for your organization.
- In addition to working with your Technical Account Manager, you can engage directly with Customer Care by creating support cases in the Google Cloud console. With Premium Support, your cases are assigned to Google technical experts, who have deep product knowledge across Google Cloud, as well as access to your architecture information and project details.
- When managing support cases as a Premium Support customer, you have access to the following features:
 - **P1 response SLO:** For Priority 1 (P1) support cases, receive the first meaningful response within 15 minutes.
 - **24/7 availability:** Receive support 24 hours a day, 7 days a week (24/7) for cases of certain priority and language.
 - **Language support:** Request support across multiple languages, including English, Japanese, Mandarin Chinese, and Korean. Depending on the case priority, language support is available 24/7 or in regional Business Hours.
 - **Case escalation:** Request additional attention on a support case by escalating the case directly from the Google Cloud console.

The customer is obligated to assist in the elimination of a malfunction/impairment. In particular, the customer shall carry out a self-check as far as possible before submitting a ticket in order to exclude the possibility that the cause of the fault lies within the customer's area of responsibility. The customer shall be obligated to notify DTIT CCoE without delay of any malfunctions, impairments of the services or damage to the equipment with a comprehensible description of the error symptoms.

Premium Support has a variable fee of 4% of the monthly charges of the customer folder.

2.2.4 Availability & Performance

Availability of GCP:

DTIT CCoE's responsibility ends at the service transfer point. The service transfer point is the router at the exit of the respective Google data center where the customer service is provided.

The availability according to the Service Level Agreement, which Google provides via the following link, applies: <https://cloud.google.com/terms/sla>

The customer can view information on system status and availability via the following link: <https://status.cloud.google.com/>

Performance

DTIT CCoE's Google Cloud Landing Zone services are based on the Google Cloud Platform functions provided by Google. DTIT CCoE's performance is therefore dependent on the performance of Google; in this respect, reference is made to the applicable Google Cloud Platform terms of service or the service level agreements described by Google. The services provided by the public cloud service provider (Google) as well as the logical and physical services within the public cloud platform, including the Hypervisor layer and the hardware devices constituting to the public cloud platform, are not part of the service provided by DTIT CCoE.

2.2.5 Maintenance

DTIT CCoE carries out regular maintenance work. If these maintenance activities lead to interruptions in the service, DTIT CCoE will inform the customer in advance. DTIT CCoE shall thereby endeavor to minimize any disruptions caused through maintenance work as far as possible.

2.3 Landing Zone Foundation

2.3.1 Organization

The service can be provided for Telekom internal users, with whom billing can take place. Currently, DTIT CCoE GCP Landing Zone is available for Deutsche Telekom IT GmbH, Magyar Telekom, and Magenta AT. Other Legal Entities within the Deutsche Telekom AG Group could also be onboarded easily.

Use of the platform for administration of the application via WebGUI and API is possible from the CN-DTAG network. The application is immediately accessible on the CN-DTAG network, but can also be configured to allow access to the application from the Internet.

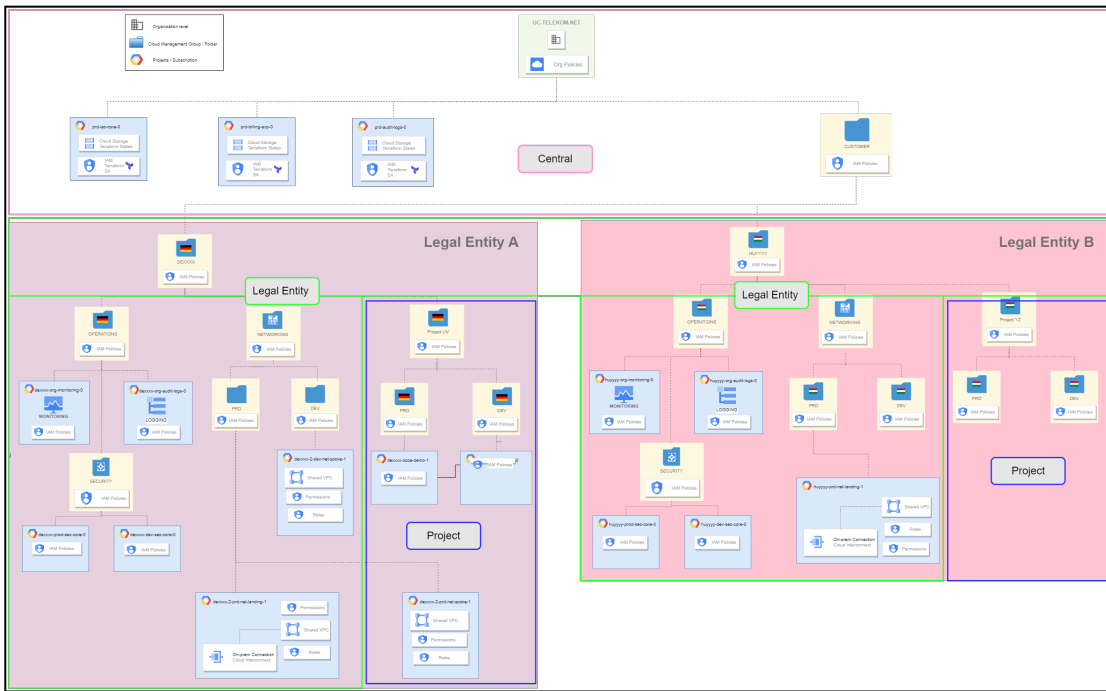
All services within the scope DTIT CCoE LZ are standardized on as-a-code solutions and provided in a fully automated manner. Required software solutions are installed in the customer's GCP-Org. The customer can create, install, update and delete its own GCP deployment manager code, as long as it does not conflict with DTIT CCoE deployment manager stacks and stack sets.

This stage within Federation Services performs two important tasks with regards to the Resource Hierarchy:

- create the top-level hierarchy of folders, and the associated resources used later on to automate each part of the hierarchy (eg. Networking)
- set organization policies on the organization, and any exception required on specific folders

The code is intentionally simple, as it's intended to provide a generic initial setup (Networking, Security, etc.), and then allow easy customization to complete the implementation of the intended hierarchy design.

The following diagram is a high level reference of the resources created and managed here.



Level	
Central	Core of the Landings Zone owned & operated from DTIT CCoE
Legal Entity (National Company/ Landesgesellschaft)	Realized with a separate Folder for each Legal Entity, because folders can contain projects and other folders, each folder could then include other sub-folders, to represent different teams/project.
Project	A project is required to use Google Cloud, and forms the basis for creating, enabling, and using all Google Cloud services, managing APIs, enabling billing, adding and removing collaborators, and managing permissions.

2.3.2 Domain Verification

gc.telekom.net is the primary domain for Cloud Identity and is verified.

To ensure a consistent and straightforward login process to the DTIT CCoE Google Cloud Landing Zone the primary email domain of the legal entity will be registered via the Google domain verification process as the secondary domain. As part of the onboarding process it will be checked if the domain is available or if your organization already created a managed Google account with your domain. Further Details you can find under [2.4.2 Identity Connect](#).

2.3.3 Customer Folder

The Customer Folder is the central core of the DTIT CCoE GCP Landing Zone Foundation. Here, each Legal Entity gets its individual area to map its project organization tailored to its needs, as well as to configure an individual on premise connection. Furthermore, necessary security controls can be adjusted on this level (e.g. country specific). All features are described in detail under [2.4 GCP Landing Zone Foundation Customer Folder](#)

2.3.4 Security Controls

GCLZ Foundation Services sets up security resources and configurations which impacts the whole organization, or are shared across the hierarchy to other projects and teams.

2.3.4.1 Cloud Identity

Cloud Identity is an Identity as a Service (IDaaS) solution that centrally manages users and groups. The DTIT CCoE Google Cloud Landing Zone Cloud Identity is federated to the Azure AD Mesh from the DTAG. Users and groups are created and managed in the AD and then synchronized to the Google Cloud Identity. You can then use Identity and Access Management (IAM) to manage access to Google Cloud resources for each Cloud Identity account.

2.3.4.2 IAM Operating

- Policies, roles and GCP groups
DTIT CCoE GCP Landing Zone provides standardized roles, policies and GCP groups for creating users in self-service to the customer. The roles are assigned to the Cloud Identity Groups (GCP groups) by DTIT CCoE. During the onboarding workshop an initial set roles, policies and groups needed by the customer will be aligned (see also [2.4.3 Access & Authentication](#))
- Password policy for GCP user accounts
DTIT CCoE GCP Landing Zone uses the password policy specified in the Azure AD Mesh to ensure corporate compliance.
- Multi-factor authentication (MFA)
DTIT CCoE GCP Landing Zone makes GCP platform multi-factor authentication (MFA) mandatory for all GCP user accounts. The GCP Landing Zone uses the multi-factor authentication (MFA) of DTAG's Azure AD Mesh, follows its rules and thus ensures compliance.

2.3.4.3 Cloud KMS

The Cloud KMS configuration allows defining keys by name (typically matching the downstream service that uses them) in different locations, either based on a common default or a per-key setting. It then takes care internally of provisioning the relevant keys and creating keys in the appropriate location.

2.3.4.4 Organizational Policy Management

This hierarchical organization of resources enables you to set access control policies and configuration settings on a parent resource, and the policies and Identity and Access Management (IAM) settings are inherited by the child resources. You can set Cloud Identity and Access Management (Cloud IAM) policies at different levels of the resource hierarchy and the resources inherit the policies applied at the parent level. Each resource has exactly one parent.

GCP Landing Zone foundation provides basic platform policies on the Top Level Org.

The following [Organizational Policies](#) are pre-configured.

Platform policies can override the Org level policies on the customer folder level, but this has to be approved by security team of the depending Customer/Legal Entity.

The setup of Google Cloud Landing Zone for DT Group follows the principle of “[shared responsibility](#)”!

2.3.4.5 Org Admins (Super Admins)

To configure or manage the Google Cloud Organization resource, a limited number of super administrator accounts are configured in the cloud identity of the DTIT CCoE GCP Landing Zone. Super Admin accounts have irrevocable administrator privileges that Google doesn't recommend for day-to-day management of your organization. For this reason, only a limited amount of admins of the DTIT CCoE GCP Landing Zone receives a separate admin account, which is uniquely identified by its own subdomain suffix [@admin.gc.telekom.net](#).

This has two advantages

1. Super Admin need to log in with a different account if you need to configure the core of the landing zone
2. A unique identification is guaranteed by the suffix and also possible with programmatic mechanisms (e.g. regex)

All activities (including disabling logging) of a super admin are logged and sent directly to the external SIEM, so that manipulation by a super admin cannot be hidden.

2.3.7 Basic Logging & Monitoring

The GCP LZ Logging and Monitoring provides different functions and tools for the GCP LZ team as well as for the foundation customers. This document is primary concerned with GCP LZ services provided for the foundation customers. This is currently planned to provide:

- [Best practices and guidelines documentation for infrastructure, OS, middleware and application monitoring](#)
- [Reference implementation](#)
- [Rich library of terraforms modules for instant monitoring](#)

2.3.7.3 Audit Logs of Administrative Activities

Audit log's purpose is to record who did what and when inside the Google Cloud Platform.

Cloud Audit Logs in GCP supports the following type of logs for Organization, Folder and Project levels:

- Admin Activity audit logs
- Data Access audit logs
- System Event audit logs
- Policy Denied audit logs

With an exclusion: publicly available resources (shared via IAM users "allUsers" and "allAuthenticatedUsers") does not generate audit logs.

Furthermore the list of services which produce audit logs are not complete but growing and can be found at <https://cloud.google.com/logging/docs/audit/services>.

Admin Activity audit logs are always written; you can't configure, exclude, or disable them. Even if you disable the Cloud Logging API, Admin Activity audit logs are still generated.

2.3.7.4 Security and Event Management (SIEM)

The Management Landing Zone is integrated with the central SIEM system, provided, and monitored by Deutsche Telekom T-Sec GmbH. Each Customer Folder must also be connected to a SIEM System. Details are described in the Customer Folder Section [2.4.9 Customer Security and Event Management \(SIEM\)](#)

2.3.8 Compliance

As part of the GC LZ Foundation Services, DTIT CCoE provides a secure Google Landing Zone, including an approved privacy and security assessments (PSA Reference No. SYS202224574) as well as a set of standard security services, with Google and Deutsche Telekom AG best practices and standards. Nevertheless, each customer folder must be subjected to a separate PSA process of the respective legal entity (see [2.4.8 Privacy Security Assessment](#)).

Additional to the Landing Zone, the DTIT CCoE offers PSA pre-approved Google services (e.g. Compute Engine) that ensured a secure deployment as well as a rapid project enabling. You can find a List of the pre-approved Services [here](#). The List will be constantly expanded so please check regularly.

2.4 GCP Landing Zone Foundation Customer Folder

2.4.1 Customer Onboarding

2.4.1.1 Affiliate Agreement & Billing Account

First Contact Point for any new customer in order to request the Google Cloud Landing Zone Foundation Services is GCLZ-Request@telekom.de -> This request FMB is dedicated to new inquiries and Project Requests to start any Onboarding Process.

One of the first steps of this process is to ensure, that the customer gets advantage of DTIT 's strong contractual partnership with Google and take part in the Cloud Master Agreement (CMA), this participation comes with the Affiliate Agreement and compliant Payment and Billing Structures as part of the GCLZ Foundation.

2.4.1.1.1 Affiliate Agreement

Based on a questionnaire, DTIT CCoE conducts an on-boarding workshop with the customer to agree on relevant information for Affiliate Agreement, the implementation of the GCLZ as well as to define the necessary guidelines for operation. The workshop takes [place online via video conference](#). On request the workshop can be held onsite as well. DTIT CCoE shall suggest a date to the customer for the on-boarding workshop.

DTIT CCoE sets up the Affiliate Agreements, aligns the (Sub) Billing Admin within your organization, based on the results of the on-boarding workshop. For this purpose, DTIT CCoE creates a new GCP-Org for the customer and sets up a folder structure (folders) and, if required, projects within this GCP-Org and installs the necessary software components for Stage1 of the Product.

With Cloud Shepherd CCoE provides a self-service order portal (with Product Stage 2) to initially create your own projects on GCP. Access is via the EMEA1/EMEA2 or ZAM account. A PSP element is required for the creation of a project, which will be bounded to your resources and be charged to. The creation of the project is immediate and can be used directly

DTIT CCoE names technical contact person for the customer and sends it a template for incident notification by e-mail, furthermore DTIT CCoE informs the customer by e-mail about completion of

the provision. The provision of services is completed when the e-mail is sent, at the latest, when the services are used.

2.4.1.1.2 Payment ID and Billing Account ID Creation

A **Cloud Billing account** is set up in Google Cloud and is used to define who pays for a given set of Google Cloud resources. Access control to a Cloud Billing account is established by IAM roles. A Cloud Billing account is connected to a Google payments profile. Your Google payments profile includes a payment instrument to which costs are charged.

A Cloud Billing account:

- Is a cloud-level resource managed in the Cloud Console.
- Tracks all of the costs (charges and usage credits) incurred by your Google Cloud usage
 - A Cloud Billing account can be linked to one or more projects.
 - Project usage is charged to the linked Cloud Billing account.
- Results in a single invoice per Cloud Billing account
- Operates in a single currency
- Defines who pays for a given set of resources
- Is connected to a Google Payments Profile, which includes a payment instrument, defining how you pay for your charges
- Has billing-specific roles and permissions to control accessing and modifying billing-related functions (established by IAM roles)

A Google Payments Profile:

- Is a Google-level resource managed at payments.google.com.
- Connects to ALL of your Google services (such as Google Ads, Google Cloud, and Fi phone service).
- Processes payments for ALL Google services (not just Google Cloud).
- Stores information like name, address, and tax ID (when required legally) of who is responsible for the profile.
- Stores your various payment instruments (credit cards, debit cards, bank accounts, and other payment methods you've used to buy through Google in the past.)
- Functions as a document center, where you can view invoices, payment history, and so on.
- Controls who can view and receive invoices for your various Cloud Billing accounts and products.

For more details about Cloud Billing account & payments profile, see https://cloud.google.com/billing/docs/concepts#billing_account

2.4.1.1.3 (Sub) Billing Admin

The Billing Account Administrator is an owner role for a billing account, and can be set on the Billing Account or on the Organization level. Billing Account Administrators can manage payment instruments, configure billing exports, view cost information, link and unlink projects and manage other user roles on the billing account.

Within the Onboarding Process we are assigning multiple **Billing Account Administrators** responsible for the customers legal entity to each of your Billing Account =Sub Billing Account Admin

With regards to the structure of Billing Accounts, that means, one for master DTIT CCoE Organization and sub-billing accounts for each new organization or one sub-billing account for project folder

For example one structure would be like this:

- MASTER DTIT ORG Billing ACCOUNT
 - Sub-Billing Account Organization - Legal Entity X

- Sub-Billing Account Organization - Legal Entity Y
- Sub-Billing Account Project - Project Legal Entity XY (i.e Project hosted in the Master Organization or inside of other organization)

The customer provides DTIT CCoE with all the necessary information, particularly a valid address for invoice, redundant Billing Account Admin contacts, a contact person authorized to make decisions and a qualified contact person. The customer ensures that all information provided is complete and accurate and that it is authorized to provide the relevant information. The customer shall always keep its information up to date.

2.4.1.2 Onboarding workshop

DTIT CCoE conducts a one-time, web-based workshop (onboarding) of up to 4 hours with the customer. DTIT CCoE explains the most important functions of the Google Cloud Platform and the customer Folder.

The subject of the workshop is:

- Introduction to user and role management
 - Roles and rights in the Google Cloud Console
- Defining Parameters of the customer Folder
 - Which Environments are needed (Prod, Dev,QA)
 - Needed Roles & Groups of the Customer Folder
- Design and coordination of the technical parameters for an on-premise connection to be created.

DTIT CCoE will send the access data for the onboarding workshop by e-mail after mutually agreeing on the date. The date of the onboarding workshop can be postponed a maximum of once. The customer is obliged to attend the on-boarding workshop on the date proposed by DTIT CCoE. As long as the on-boarding workshop takes place at customer's location, the customer provides the necessary operating and working equipment as well as workspace. The customer shares the information required for implementing the customer environment based on the questionnaire with DTIT CCoE.

DTIT CCoE sets up the Customer Folder based on the results of the onboarding workshop. DTIT CCoE informs the customer by e-mail about completion of the provision. The provision of services is completed when the e-mail is sent, at the latest, when the services are used.

2.4.2 Identity Connect

Identity and Access Management (IAM) including Roles & Policies is implemented for GCP LZ Foundation Services. This also includes a Standard Blueprint for Single Sign On (SSO) by using the Email Address. To achieve this, it is necessary to verify the customer's primary email domain (e.g. [affiliate.com](#)) with Google (see also [2.3.2 Domain Verification](#))

The customer allows to register the primary email domain of the legal entity at Google via the Google domain verification process as the secondary domain. If the domain is occupied at Google, the Domain Contestation Process can be started in order to claim the domain without major impact on the existing services the domain is registered for. After successful domain verification/registration, the Google Cloud Landing Zone cloud identity can be used and connected to an on-premises or Azure AD. The customer supports this process with all necessary information and contacts needed.

2.4.3 Access & Authentication

During the onboarding workshop, the initial authorization groups for the operation of the client folder and any projects are agreed and created during technical onboarding. After the customer folder has been transferred to the customer's company, it is the customer's responsibility to maintain the authorization groups and, if necessary, to adapt or expand them to meet current needs. The assignment of authorizations should always follow the principles of least privilege. Furthermore, the customer is also responsible for ensuring that the authorizations of employees who have left the company are removed in a timely manner.

2.4.4 Network

During the onboarding workshop, the customer is presented with various options for network management and a joint decision is made on how the individual network configuration for the customer's folder will be created. For the operation of the customer folder, one Classless Inter-Domain Routing (CIDR) block is required in the minimum. For example, 10.0.0.0/16.

Furthermore, it must be decided whether and in what form a connection to the local network of the respective legal entity is necessary. After the initial creation of the network configuration during the initial creation of the customer folder, the operation is handed over to the respective operating team of the legal entity. This team is then also responsible for future adjustments or extensions to the network.

2.4.5 Security

The customer is responsible for the security of his customer folder. Thus, he is obliged to protect the customer folder, the GCP projects contained therein, the operating systems and applications and data within his VMs from attacks by third parties and misuse and to keep them free of malware.

In addition to the central security policies inherited from the landing zone, local extensions may be necessary for this.

 Although the customer is able to change the inherited policies on folder level this is forbidden and the policies will be re-activated automatically.

All connections to and from the internet must be logged by proxy and, if possible, all relevant network traffic (VPC flowlogs). To achieve this, the use of a forward and/or reverse proxy is necessary.

Depending on the type of possible Om-Prem connection, it may be necessary to encrypt the data transfer, as it is not natively implemented in all connection types (z.b. Google Cloud Interconnect)

2.4.6 Backup & Disaster Recovery

Central Backup is not implemented for the Landing Zone itself.

The entire Landing Zone is created as Infrastructure as Code, deployed with the Magenta CI/CD Pipeline. Magenta CICD is backed up each night and additionally around 30 minutes before each version update of GitLab, a backup is manually triggered.

Customer Obligation:

All Projects are responsible to ensure the Security also in regards of Proper Backups of their Resources. For Example with managed backup and disaster recovery (DR), Google offers a service

for centralized, application-consistent data protection. Protect workloads running in Google Cloud and on-premises by backing them up to Google Cloud.

 Google Cloud offers different Solutions to protect your Application and Data. Please consult the [Google Documentation](#) to get an Overview about the possibilities.

2.4.7 Patch management

Each customer must ensure that its services and applications are regularly updated with the latest security patches, unless they are PaaS or SaaS services from Google and Google is therefore responsible for the constant updating of the services.

2.4.8 Privacy Security Assessment (PSA)

Even though the Landing Zone is basically PSA approved, each customer must go through a separate PSA for their project/application.

If the central SIEM service will be used by customer, he can use the Landing Zone approval as basic for his approval process.

All used GCP Services should be listed as [SOC2](#) or [C5](#) compliant on the Google Pages. If this is not the case they are also usable but have to pass an extended PSA process.

2.4.9 Customer Security and Event Management (SIEM)

The customer is obliged to monitor his customer folder and all therein contained Resources with a SIEM system for possible cyber-attacks, protecting against potential production downtime, data loss, image damage, etc. If the customer does not have its own SIEM or does not wish to set up and operate one itself, it is possible to connect the customer folder to Deutsche Telekom T-Sec GmbH central SIEM system. If customer decides to use the central SIEM service from Deutsche Telekom T-Sec GmbH for security monitoring, it has a one-time onboarding cost, and a monthly operation cost. Also in this case the customer designates a team responsible to react to the queries from the Cyber Defence Center in the event of security threats, fault reports or service requests and ensures that this team can be reached 24 by 7.

2.4.10 Customer Folder Operations

After the initial creation during the customer onboarding, the customer is responsible for the operation and maintenance of the customer folder and its content. This also includes the creation of further projects including the necessary assignment of authorizations.

Adjustments to the network and adjustments to necessary local compliance regulations (security controls) are to be carried out independently. The DTIT CCoE team is happy to provide consulting support.

The customer provides a qualified and authorized contact person for all topics regarding Customer Folder. The customer has extensive rights on the system side, which allow a change in the basic settings of all resources in the GCP-Org. Notwithstanding this, the customer is obliged to make no changes to the resources required for the provision of GCP LZ Foundation services.

2.4.11 Usage

The customer is responsible for its own capacity management and reduces or expands its project resources independently.

2.4.12 Delete at end of use

Upon termination of services, all customer access will be deactivated and all accounts, tenant projects will be deleted. The customer must independently and on his own responsibility back up his data via download before completing the service.

Change history

There may be changes to this OLA. An officially modified version can be uniquely identified by the version number.

Version	Last revised	Editor	Changes/ comments
1.0	 07.03.2023	Carsten Fraszczak	Published

Glossary

Term	Description
AI	Artificial Intelligence
API	Application Programming Interface (typically used for automatic control and/or integration with higher-level orchestration)
AUP	Acceptable Use Policy
AZ	Availability Zone
CET/CEST	Central European Time/ Central European Summer Time
CMEK	Customer Managed Encryption Keys
DB	Database
DR	Disaster Recovery (protection from a data center outage, e.g. catastrophes)
GCP	Google Cloud Platform
GB	Gigabyte

Term	Description
GPU	Graphics Processing Unit or graphics processor that supports intensive computing operations
GUI	Graphical User Interface
IaC	Infrastructure as Code
IAM	Identity and Access Management (user management)
IaaS	Infrastructure as a Service
IP	Internet Protocol
IPC	inter-Process Communication
KMS	Key Management Service
MCS	Managed Cloud Services
MESZ	Central European Summer Time
ML	Machine Learning
OLA	Operational Level Agreement
OS	Operating System
PSA	Privacy and Security Assessment. Security and data protection policies of Telekom
Reaction Time Service Desk	The maximum time between reporting an incident or request by the customer and first-time confirmation of the current status by Telekom
SIEM	Security and Event Management
SLA	Service Level Agreement
TB	Terabyte

Term	Description
VM	Virtual machine
VPN	Virtual Private Network (typically between IPsec and Site2Site scenario) – enables an encrypted communication via unencrypted connection ways like Internet

2. OLA Managed Project



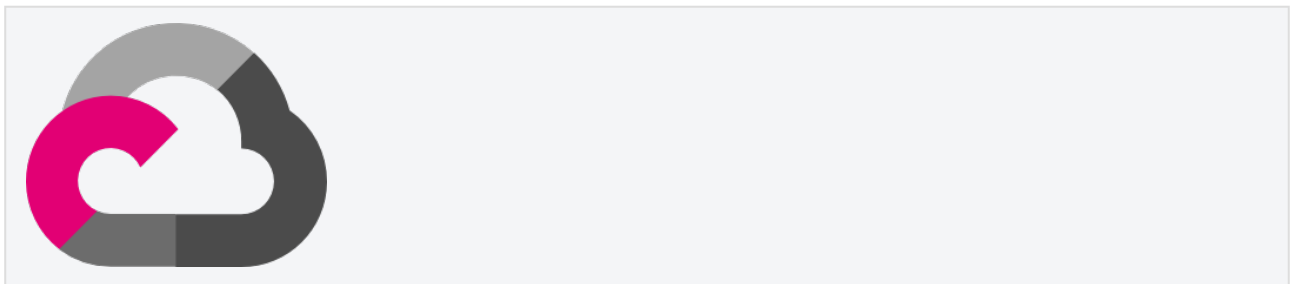
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1 Introduction

With the Google Cloud Platform (GCP) Landing Zone, the ‘Cloud center of excellence’ (CCOE) within Deutsche Telekom IT provides a secure Infrastructure as Code (IaC) environment on the Google Cloud Platform (GCP), for the operation of DTIT applications and workloads as a service. It supports compliance with the corporate requirements, regarding data privacy and security and follow GCP best practices. Among other use cases, the solution focuses on Google Analytics Solutions for BI and data driven businesses with open-source-affinity. The use of the Landing Zone is highly automated, especially for Account and Project Creation by using a Self-Service Portal & automated cost distribution into DTAG.



Mission/Vision

Our Motivation to establish Google as an alternative Hyperscaler at Deutsche Telekom AG for mainly Cloud Native, Data Analytics BI & AI Use Cases is, that we strongly believe in enabling business by adding this value throughout the entire group.

2 Scope of Essentials Service

With the DTIT CCoE GCP Landing Zone, Google is responsible for providing basic protection of the infrastructure offered and running through Google Cloud Platform (GCP). This infrastructure consists of the hardware, software, network, and facilities on which and in which Google Cloud Services run.

The Landing Zone provides functionalities on several levels:

Provisioning of GCP accounts/environments for projects/applications according to their needs (e.g. different environments like development or production, connectivity requirements etc.)

Identity provisioning, most importantly federation with the DTAG Active Directory to authorize access to GCP accounts.

Central logging and monitoring of LZ management (CI/CD, provisioning, billing, security management, etc..) with recording of, among others, all GCP API activities on LZ Management Level in the landing zone and providing means to protect and analyze that data while ensuring privacy and security requirements.

Note that the DTIT CCoE GCP Landing Zone does not contain servers (neither virtual machines nor bare metal). Rather it consists of resources such as accounts, VPCs, IAM roles, Compute Engine, Cloud Firewall, Cloud Load Balancing, Pub/Sub topics or Cloud Storage buckets, to name just a few, leveraging the respective fully-managed services from GCP. By using this environment, the DTAG applications can go-live on GCP faster, are running in an approved environment compliant with all requirements and, most importantly, application teams can focus on delivering business value as they are being freed from heavy lifting regarding connectivity, billing or implementing the required controls.

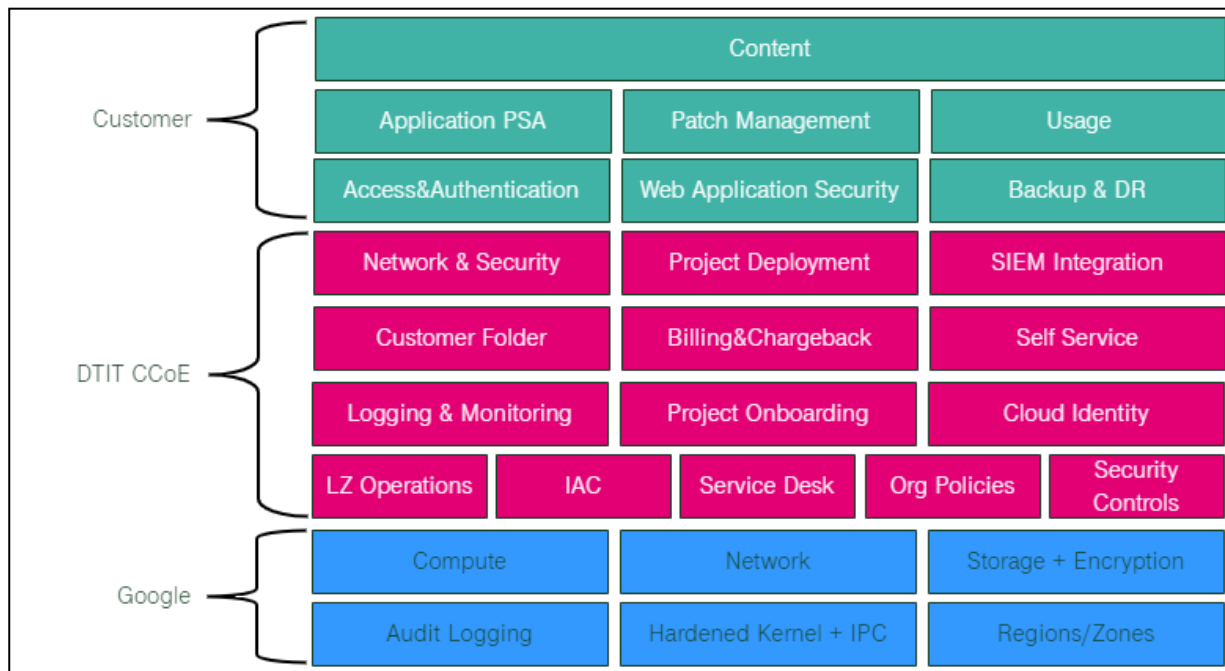
Scope of the document:

The GCP Landing Zone Managed Project provides a professional managed service that provides customers with a Google project through a self-service portal, enabling them to build applications that are configured in line with security policies, compliance policies, and cloud-native best practices. Scope of this document is to describe the Managed Service with appropriate level of details.

If the customer requirements go beyond a project deployment and the Status of a Legal Entity is given, the "[GCP Landing Zone Foundation](#)" of DTIT CCoE may be the right choice.

2.1 Shared Responsibility Model (Managed Project)

The shared responsibility model regulates the responsibilities and accesses of the different parties within the Google Cloud Landing Zone of the DTIT.



Google

Google is responsible for the security of the infrastructure on which all services offered by Google Cloud are running. This infrastructure consists out of hardware, software, network as well as the Datacenter on and in which the Google Cloud services are running.

DTIT CCoE

The DTIT offers with the Google Cloud Service Managed Project the basis for rapid project onboarding to support customer projects to the greatest extent possible. In detail, it includes the following building blocks:

- Project Onboarding - via Cloud Shepherd Service Portal can be a Managed Project with access to the CN-DTAG Network be ordered
- Landing Zone Operations - The landing zone operation includes the constant maintenance and care of the landing zone both at the code base and in particular at the continuous adjustment of the security parameters to the current threat situation.
- Cloud Identity - Cloud Identity is used to manage Google Cloud Platform users from a central location in the Google Admin console (admin.google.com). DTIT CCoE provides the maintenance and care of this system as well as the connection of the local identity providers.
- Logging & Monitoring - The Google Landing Zone is monitored at the top level to detect security incidents at a central location to be able to react quickly. The customers are responsible for the monitoring, logging and security management of their applications.

Customer

With the delivery of the managed Project, the Project Owner take over the Responsibility for the following Topics:

- WebApplication Security - It is a security best practice and a must when using public cloud to follow the principle of least privilege when creating IAM policies to grant entities access to the Google Cloud.

- Application PSA - By requesting a managed Project, the obligation for PSA as well as Worker's Council involvement is acknowledged.
- Access & Authentication - Please have in mind that you are responsible for the management of those users who are granted access to the self-service portals and your virtual machines. Please review carefully and regularly who is granted access to.
- Backups & DR - By default your data and cloud resources will not be backed up and cannot be recovered within the self-service portal. Please ensure that you possess suitable tools and back up your data by yourself.
- Patch management - Please ensure that your operating systems as well as applications are always up to date. In addition, make use of the solution's flexibility to define when your organization can tolerate potential patch-related outages best.
- Usage - In general, customers are responsible for the capacity management and with this the cost of their Environment. Cost management is the key to successful cloud infrastructure operation.
- Order data processing – Please survey carefully if you process personal data on your GCP Environment.

2.2 Landing Zone Operations by DTIT CCoE

2.2.1 Service Desk

DTIT CCoE support services are provided within the European Union. The GCP services itself are available worldwide according to the customer's choice.

DTIT CCoE Support is the central contact partner for the customer's key users regarding incidents and general service requests. Support is available to the customer workdays to the OFFICE TIME and can be contacted at the following e-mail address and telephone number:

- A ticket can be raised on <https://jira.telekom.de/servicedesk/customer/portal/301/group/2898>
- alternatively via e-mail GCLZ-Support@telekom.de -> connected to our Ticket system for Service Request

2.2.2 Operating times

- Operating times (Uptime) refers to the period during which the GCP service is available. Uptime is 7 x 24 hours.
- DTIT CCoE Attended Operation Time (AOT) is the period during which technical support is available and during which incidents affecting system availability are handled. The AOT period is the OFFICE TIME. The OFFICE TIME is Mondays to Fridays, excluding national holidays (Germany), from 09:00 to 16:00
- Response time 8 h Within the service time zone

2.2.3 Google Support

The DTIT CCoE Cloud Master Agreement (CMA) ensures [Google Premium Support Level](#). Premium Support is a paid support offering designed for enterprises that run mission critical workloads and require fast response times, platform stability, and increased operational efficiencies. Within the Affiliate Agreement, customer are able to take advantage by joining this support level. Premium Support will be invoiced on a consumption based, shared cost model. If the customer is in a

permission group higher than Viewer in a project or folder, he has the possibility to open a support case in the [Google Console](#) for his project.

- As a Premium Support customer, you are assigned a shared, named DTAG Technical Account Manager (TAM). Technical Account Managers are trusted technical advisors that focus on operational rigor, platform health, and architectural stability for your organization.
- In addition to working with your Technical Account Manager, you can engage directly with Customer Care by creating support cases in the Google Cloud console. With Premium Support, your cases are assigned to Google technical experts, who have deep product knowledge across Google Cloud, as well as access to your architecture information and project details.
- When managing support cases as a Premium Support customer, you have access to the following features:
 - **P1 response SLO:** For Priority 1 (P1) support cases, receive the first meaningful response within 15 minutes.
 - **24/7 availability:** Receive support 24 hours a day, 7 days a week (24/7) for cases of certain priority and language.
 - **Language support:** Request support across multiple languages, including English, Japanese, Mandarin Chinese, and Korean. Depending on the case priority, language support is available 24/7 or in regional Business Hours.
 - **Case escalation:** Request additional attention on a support case by escalating the case directly from the Google Cloud console.

The customer is obligated to assist in the elimination of a malfunction/impairment. In particular, the customer shall carry out a self-check as far as possible before submitting a ticket in order to exclude the possibility that the cause of the fault lies within the customer's area of responsibility. The customer shall be obligated to notify DTIT CCoE without delay of any malfunctions, impairments of the services or damage to the equipment with a comprehensible description of the error symptoms.

Premium Support has a variable fee of 4% of the monthly charges of your project.

General: Google Support assists with all technical questions/problems with Google Services. For all other questions, e.g. access rights or billing questions, please contact DTIT CCoE Support.

2.2.4 Availability & Performance

Availability of GCP:

DTIT CCoE's responsibility ends at the service transfer point. The power transfer point is the exit point from the Google data center to the Internet.

The availability according to the Service Level Agreement, which Google provides via the following link, applies: <https://cloud.google.com/terms/sla>

The customer can view information on system status and availability via the following link: <https://status.cloud.google.com/>

Performance

DTIT CCoE's Google Cloud Landing Zone services are based on the Google Cloud Platform functions provided by Google. DTIT CCoE's performance is therefore dependent on the performance of Google; in this respect, reference is made to the applicable Google Cloud Platform terms of service or the service level agreements described by Google. The services provided by the public cloud service provider (Google) as well as the logical and physical services within the public cloud

platform, including the Hypervisor layer and the hardware devices constituting to the public cloud platform, are not part of the service provided by DTIT CCoE.

2.2.5 Maintenance

DTIT CCoE carries out regular maintenance work. If these maintenance activities lead to interruptions in the service, DTIT CCoE will inform the customer in advance. DTIT CCoE shall thereby endeavor to minimize any disruptions caused through maintenance work as far as possible.

2.2.6 Security Controls

The Google Cloud Landing Zone sets up security resources and configurations which impact the whole organization, or are shared across the hierarchy to other projects and teams.

2.3.6.1 Cloud Identity

Cloud Identity is an Identity as a Service (IDaaS) solution that centrally manages users and groups. The DTIT CCoE Google Cloud Landing Zone Cloud Identity is federated to the Azure AD Mesh of the DTAG. Users and groups are created and managed in the AD and then synchronized to the Google Cloud Identity. You can then use Identity and Access Management (IAM) to manage access to Google Cloud resources for each Cloud Identity account.

2.3.6.2 IAM Operating

- Policies, roles and GCP groups
DTIT CCoE GCP Landing Zone provides standardized roles, policies and GCP groups for creating users in self-service to the customer. The roles are assigned to the Cloud Identity Groups (GCP groups) by DTIT CCoE.
- Password policy for GCP user accounts
DTIT CCoE GCP Landing Zone uses the password policy specified in the Azure AD Mesh to ensure corporate compliance.
- Multi-factor authentication (MFA)
DTIT CCoE GCP Landing Zone makes GCP platform multi-factor authentication (MFA) mandatory for all GCP user accounts. The GCP Landing Zone uses the multi-factor authentication (MFA) of DTAG's Azure AD Mesh, follows its rules and thus ensures compliance.

IMPORTANT NOTE: Creation and changes to IAM rights and roles inside the Google Cloud Console are only allowed from inside telekom networks.

2.3.6.3 Cloud KMS

The Cloud KMS configuration allows defining keys by name (typically matching the downstream service that uses them) in different locations, either based on a common default or a per-key setting. It then takes care internally of provisioning the relevant keys and creating keys in the appropriate location.

2.3.6.4 Organizational Policy Management

This hierarchical organization of resources enables you to set access control policies and configuration settings on a parent resource, and the policies and Identity and Access Management (IAM) settings are inherited by the child resources. You can set Cloud Identity and Access

Management (Cloud IAM) policies at different levels of the resource hierarchy and the resources inherit the policies applied at the parent level. Each resource has exactly one parent.

DTIT CCoE GCP Landing Zone provides basic platform policies on the Top Level Org.

The following [Organizational Policies](#) are configured.

The setup of Google Cloud Landing Zone for DT Group follows the principle of “[shared responsibility](#)”!

2.3.6.5 Org Admins (Super Admins)

To configure or manage the Google Cloud Organization resource, a limited number of super administrator accounts are configured in the cloud identity of the DTIT CCoE GCP Landing Zone. Super Admin accounts have irrevocable administrator privileges that Google doesn't recommend for day-to-day management of your organization. For this reason, only a limited amount of admins of the DTIT CCoE GCP Landing Zone receives a separate admin account, which is uniquely identified by its own subdomain suffix [@admin.gc.telekom.net](#).

This has two advantages

1. Super Admin need to log in with a different account if you need to configure the core of the landing zone
2. A unique identification is guaranteed by the suffix and also possible with programmatic mechanisms (e.g. regex)

All activities (including disabling logging) of a super admin are logged and sent directly to the external SIEM, so that manipulation by a super admin cannot be hidden.

2.2.7 Logging & Monitoring

The GCP LZ Logging and Monitoring provides different functions and tools for the GCP LZ team as well as for the managed project customers.

2.2.7.1 Audit Logs of Administrative Activities

Audit log's purpose is to record who did what and when inside the Google Cloud Platform.

Cloud Audit Logs in GCP supports the following type of logs for Organization, Folder and Project levels:

- Admin Activity audit logs
- Data Access audit logs
- System Event audit logs
- Policy Denied audit logs

With an exclusion: publicly available resources (shared via IAM users "allUsers" and "allAuthenticatedUsers") does not generate audit logs. Because for those Accesstypes it is not trackable who has accessed it.

Furthermore the list of services which produce audit logs are not complete but growing and can be found at <https://cloud.google.com/logging/docs/audit/services>.

Admin Activity audit logs are always written; you can't configure, exclude, or disable them. Even if you disable the Cloud Logging API, Admin Activity audit logs are still generated.

2.2.7.2 Security and Event Management (SIEM)

The Management Landing Zone is integrated with the central SIEM system, provided, and monitored by Deutsche Telekom T-Sec GmbH. All the Audit Logs in the Org main Bucket and also their forwarded Versions in the PubSub and Cloud Storage Bucket have a retention of 90 days.

Note that the W5BaseID is required/used for figuring out the related incident assignment group for the managed Project. So, every managed Project to have a corresponding application entry in DARWIN (DTIT CMDB). In order to get your application registered into DARWIN please register your application at <https://darwin.telekom.de>. Should you have any troubles in doing so, please contact the Google Cloud Landing Zone Team via Support Ticket. If you already have a DARWIN entry feel free to use it or to issue a separate one for your Google Cloud implementation. Note that Google Managed Projects will not be registered in Darwin automatically.

Critical security issues are forwarded to Cyber Defense Center (CDC) which creates a ticket for the incident assignment group given in the Darwin entry of the respective Managed Project; also other remediating actions might be applied such as isolation of suspicious instance or user deactivation.

Further information on all the controls and cloud security products can be found at the YAM page of DTIT's DevSecOps team <https://yam-united.telekom.com/pages/devsecops/apps/content/home>. In case of any questions about the controls you can contact devsecops@telekom.de.

2.2.8 Compliance

As part of the GC LZ Managed Project, DTIT CCoE provides a secure Google Landing Zone, including an approved privacy and security assessments (PSA Reference No. SYS202224574) as well as a set of standard security services, with Google and Deutsche Telekom AG best practices and standards. Nevertheless, each Application and/or Project must be subjected to a separate PSA process.

Additional to the Landing Zone, the DTIT CCoE offers PSA pre-approved Google services (e.g. Compute Engine) that ensured a secure deployment as well as a rapid project enabling. You can find a List of the pre-approved Services [here](#). The List will be constantly expanded so please check regularly.

2.2.9 Workers' Council participation

In addition to the application specific PSA approval process Of course, Workers' Council representative participation ("Mitbestimmung") also needs to be ensured. In doing so, relevant participation items such as security and data privacy, or application accessibility ("Barrierefreiheit") need to be addressed.

2.3 Landing Zone Managed Project

2.3.1 Project Administration (CS)

With [Cloud Shepherd](#), application teams can create their cloud environments themselves and get a new environment within minutes. Only a minimal set of information is required to get started. Within Cloud Shepherd you can create a new Google cloud project by deciding on a name, entering a cost center and providing a list of users with full administrative access. Right after the approval the environment is created within 5 minutes and can be accessed with your regular logon credentials.

With the Version 1.0 of the Cloud Landing Zone this Service is only for the DTIT available. Other German Subsidiaries will come later this year.

2.3.1.1 Project Creation

For the preparation of a project, the following information is required at least:

- Name of the project
- Cost Element
- Environment (dev or prod)
- Project Members

After completion of the project through the Self Service Portal, the client (Project Owner) will be informed by e-mail. In addition to the name of the project, the mail also contains a link with a direct access to the Google Console.

2.3.1.2 Project Access Management

With the creation of the Project, two authorization groups are created to regulate access to the project.

- Project Owner
- Project Members

The members of these groups can be managed or customized by the owner on the groups.google.com page.

Correspondingly, these groups have the following roles in Google Identity Access Management (IAM):

- Project Owner = [Owner Role](#)
- Project Members = [Editor Role](#)

The Project Owner is responsible for configuring all further permissions in a fine-grained manner according to the least-privilege principles for their application as required. If there are more groups (e.g. for Applications) needed, this has to be done in the Azure Active Directory first.

2.3.1.3 Project Member

To access the Managed Project service, an account in Deutsche Telekom's Azure Active Directory is required. These accounts are replicated to the Google Cloud Identity to enable working in the Google Cloud. By applying for a project and adding users, they agree to the transfer of participants' personal data. Detailed information can be found on the [MyWiki page of the Google Cloud Landing Zone](#).

External employees may first need to be created Microsoft 365 Guest Account at Deutsche Telekom. Further information can be found here <https://yam-united.telekom.com/pages/workplace-experience/apps/wiki/wiki/list/view/a0e9aced-896c-499e-9fb1-cb2d44b7e8b7>

2.3.1.4 Service Accounts

For the Service Managed Project, the use of traditional service accounts is disabled. Alternatively, you should use Workload Identity Federation.

Why?

Traditionally, applications running outside Google Cloud have used [service account keys](#) to access Google Cloud resources. Service account keys are powerful credentials and can represent a security risk if they are not managed correctly.

With identity federation, you can use Identity and Access Management (IAM) to grant external identities [IAM roles](#), including the ability to impersonate service accounts.

Configuration instructions can be found on the [MyWiki page of the Google Cloud Landing Zone](#).

If it is not technically feasible to operate the service with WIF, please contact the CCoE Service Desk to discuss an exception and the separate use of a service account.

2.3.1.5 Quotas

Additional to the Global Restriction due to the [Organizational Policy](#) there are set some Quotas to avoid abuse and/or financial damage. If one of this Ressource is needed for the Project, please reach out to the CCoE Support, so that the Quotas could be adjusted for your project.

The Full List of the current restricted Resources could be found in the [Technical Documentation](#) of this Wiki.

2.3.1.6 Sandboxes

Sandboxes are there for users who want to just try out things on an isolated environment or users who are completely new to Public Cloud environments and just want to have a deeper look into it. However, due to security requirements also sandbox projects need to be implemented similar to all other projects types while on the other hand CCoE needs to charge these projects types as well as they do not have a budget available to sponsor Sandboxes. A sandbox is no free of charge type of account.

Sandboxes are available after PI 18 earliest in September 2023

2.3.2 Regions/Zones

The DTIT CCoE Service Managed Project service is limited to Google's europe-west3 (Frankfurt, Germany) and europe-west4 (Eemshaven, Netherlands) regions. Within these regions there are no further regulations, so all zones can be used.

2.3.3 Networking

2.3.3.1 Network Connect

The connection Google Cloud to on-prem is established via a redundant Google Partner Interconnect. On the Deutsche Telekom site the InfraSelect MPLS product is used. All network traffic to and from Google is routed through an on-prem firewall cluster. The connection (CCR to Google) is not encrypted. To increase the security, only extra for the communication partners encrypted traffic is permitted. An example is HTTP encrypted with Transport Layer Security (TLS) or a Secure Shell Protocol (SSH) connection.

All Traffic from and to Google will be filtered thru a Firewall. Everything (except ICMP ping) is blocked from Google to the on-prem network. Only ICMP ping and HTTPS with TCP port 443 are allowed from on-prem to the Google Cloud. HTTP with TCP port 80 is only allowed to enable an HTTPS redirect.

Project / Application Specific Firewall Port Enabling must be discussed and approved in the application's PSA approval process with the Platform Security Manager. Also, Google Landing Zone Team must be involved.

2.3.3.2 Cloud Networking

Many services in the Google Cloud no longer require dedicated IP addresses, but are addressed via APIs. For some services (e.g. compute engines) IP addresses are still required. For this purpose, two Private Sub-Networks out of the DTIT private IP Ranges are provided in the GCP Landing Zone Managed Project. One network each in europe-west3 (Frankfurt) and europe-west4 (Netherlands) region. An IP address can be obtained from these networks during configuration of the compute engine. A fixed reservation of an IP address is also possible.

Important note: The two networks are shared networks that they share with other departments of Deutsche Telekom. There are separate networks per Telekom subsidiary, so that sharing only takes place within a subsidiary company, but not outside.

A dedicated Customer Network Area will come earliest in PI18

2.3.3.3 Private Service Endpoints

In order to be able to reach Google services and their APIs via private addresses from the CN-DTAG network, a Private Service Connect endpoint was created in the Google Cloud Landing Zone. All common Google services can be accessed via this endpoint by addressing different DNS names. A detailed overview of the services as well as setup instructions can be found in the following [Knowledge Base](#) article.

2.3.3.4 Public IP

Public IP on Virtual Machines is not allowed and will be denied automatically via Policy. Exemptions have to be discussed and approved in the application's PSA approval process with the Platform Security Manager. Also, Google Landingzone Team must be involved.

2.3.3.5 Internet Access

Internet access is divided into 2 directions outgoing and incoming. Both directions are subject to the condition that all traffic must be logged. The Logging for the GCP Landing Zone Managed Project, take place in the central Network Project.

2.3.3.5.1 Forward Proxy Area

The high security requirements of the Google Cloud Landing Zone require that all network traffic to and from the Internet must be recorded. For this reason, all requests are routed via a so-called Forward Proxy Area (FPA). This consists of a transparent proxy for the normal ports and an explicit proxy for the so-called high ports. The FPA solution has already implemented whitelists with common websites. If you need access to additional destinations on the Internet, please open a ticket in the service desk.

2.3.3.5.2 Reverse Proxy Area

The Reverse Proxy Area is required when services are to be made accessible from the Public Internet. This is not available with the production start of the Google Cloud Landing Zone, but will only be implemented in the course of PI 17.

2.3.3.6 VPC Sharing

VPC peering, i.e., the connection of two virtual private cloud (VPC) networks so that resources in each network can communicate with each other, is disabled for security reasons.

2.3.3.7 DNS Forwarding and Peering

For the GCP Landing Zone Managed Project is a private DNS zones used. Cloud DNS is enabled by GCLZ Team in every VPC. You can manage your own zones and DNS entries with Cloud DNS. By default, we create a zone for the Google API DNS names during our setup.

2.3.4 Content

The customer is responsible for his data. This concerns the processing as well as the appropriate backup of the data (see also [2.3.5 Backup & Disaster Recovery](#)). The customer shall check on his own responsibility whether the data he transmits to the Google Cloud Platform in connection with the use of the service constitute personal data and whether the processing of such personal data is permissible.

2.3.5 Privacy Security Assessment (PSA)

Even though the Landing Zone is basically PSA approved, each customer must go through a separate PSA for their project/application.

All used GCP Services should be listed as [SOC2](#) or [C5](#) compliant on the Google Pages. If this is not the case they are also usable but have to pass an extended PSA process.

2.3.6 Backup & Disaster Recovery

Central Backup is not implemented for the Landing Zone itself.

The entire Landing Zone is created as Infrastructure as Code, deployed with the Magenta CI/CD Pipeline. Magenta CI/CD is backed up each night and additionally around 30 minutes before each version update of GitLab, a backup is manually triggered.

Customer Obligation:

All Projects are responsible to ensure the Security also in regards of Proper Backups of their Resources. For Example with managed backup and disaster recovery (DR), Google offers a service for centralized, application-consistent data protection. Protect workloads running in Google Cloud and on-premises by backing them up to Google Cloud.

Backup

Google Cloud offers different Solutions to protect your Application and Data. Please consult the [Google Documentation](#) to get an Overview about the possibilities.

2.3.7 Patch management

Each customer must ensure that its services and applications are regularly updated with the latest security patches, unless they are PaaS or SaaS services from Google and Google is therefore responsible for the constant updating of the services.

2.3.8 Billing & Chargeback

Google Cloud operates on a consumption model. You are charged based on how much of a particular resource or product you use over a given payment period. Products measure consumption in different ways, for example:

- As an amount of time (how many seconds a machine was running)
- As a volume (how much data is stored)

- As the number of operations executed
- As variations of those concepts

2.3.8.1 Chargeback

The costs incurred for your project are booked to your Cost Element (given with the Project Creation) on a monthly basis. If you have any questions about your billing, please open a ticket in the service desk.

2.3.8.2 Billing Alerts

Later used for Budget alert.

CloudShepherd users are currently not able to enter the amount, there we set a fix amount:

- Dev = 1000
- Prd = 10000

At a later point in time we will develop this further if necessary.

2.3.9 Usage

Make sure you understand how the billing works for the components in your system so that you can accurately gauge your costs. Each product provides [detailed pricing information](#) in its documentation. Many products provide a [Free Tier](#) where the cost of any consumption below a certain threshold is credited to your billing account so you do not incur charges. Resources consumed beyond what is offered by the Free Tier are charged to your cost element.

For more details about Google Cloud pricing philosophies, innovations, and discounts, see the [Pricing](#) page.

2.3.10 Delete at end of use

Upon termination of services, all customer access will be deactivated the project will be deleted. The customer must independently and on his own responsibility back up his data via download before completing the service. A deleted Project and the data stored in them may be recovered for a limited time (10 Days) before the projects are permanently deleted. In case a project Restoring is necessary get in contact with the [Service Desk](#).

Change history

There may be changes to this OLA. An officially modified version can be uniquely identified by the version number.

Version	Last revised	Editor	Changes/comments
1.0	 09.06.2023	Carsten Fraszczak	Published

Glossary

Term	Description
AI	Artificial Intelligence

Term	Description
API	Application Programming Interface (typically used for automatic control and/or integration with higher-level orchestration)
AUP	Acceptable Use Policy
AZ	Availability Zone
CCR	Cloud Core Router
CET/CEST	Central European Time/ Central European Summer Time
CMEK	Customer Managed Encryption Keys
DB	Database
DR	Disaster Recovery (protection from a data center outage, e.g. catastrophes)
FPA	Forward Proxy Area
GCP	Google Cloud Platform
GB	Gigabyte
GPU	Graphics Processing Unit or graphics processor that supports intensive computing operations
GUI	Graphical User Interface
IaC	Infrastructure as Code
IAM	Identity and Access Management (user management)
IaaS	Infrastructure as a Service
IP	Internet Protocol
IPC	inter-Process Communication

Term	Description
KMS	Key Management Service
MCS	Managed Cloud Services
MESZ	Central European Summer Time
ML	Machine Learning
OLA	Operational Level Agreement
OS	Operating System
PSA	Privacy and Security Assessment. Security and data protection policies of Telekom
RPA	Reverse Proxy Area
Reaction Time Service Desk	The maximum time between reporting an incident or request by the customer and first-time confirmation of the current status by Telekom
SIEM	Security and Event Management
SLA	Service Level Agreement
TB	Terabyte
VM	Virtual machine
VPN	Virtual Private Network (typically between IPsec and Site2Site scenario) – enables an encrypted communication via unencrypted connection ways like Internet

PSA related



Underlying Links

- [Pre Approved Services](#)

Table of Contents

- [Basics](#)
- [Transfer Impact Assessment \(TIA\) DTIT](#)
 - [Simplified TIA](#)
 - [Long TIA](#)
 - [How to get the needed TIA Documents](#)
- [Transfer Impact Assessment \(TIA\) other legal entities](#)

Basics

The Landing Zone delivers a baseline of security and so PSA depending configurations, which reduces the amount of technical topics which has to be covered by the customers PSA process. Never the less there are parts of the customer solution running on GCLZ which are only be known by the customer/developer of the solution. This logically implicates, that every project has to fulfill its own PSA process.

Related Systementries:

Darwin id: GCP@DTIT(PROD)

Cape id: SPL-204: GCP@DTIT(PROD) 1.0

PSA Project Number: PSA-2022-2178

Jira@BR: DTIT-526 (Steckbrief: GCP DTIT-684)

Transfer Impact Assessment (TIA) DTIT

The TIA is part of the PSA process when data third countries-transfer is part of the solution.

Simplified TIA

For the GCLZ we can provide for customers belonging to DT-IT a simplified TIA document.

Long TIA

For every other customer the long TIA is necessary which can be found here in the [long TIA link](#)

How to get the needed TIA Documents

To get the required TIA documents, please send an email to the [SRC.BIA functional mailbox](#).

The Colleagues decide which TIA Process is the right for you Application and send you the necessary Documents.

More Information about the TIA is to find on the sites of Group Privacy:

<https://yam-united.telekom.com/pages/group-privacy/apps/wiki/wiki/list/view/07fab617-0deb-4030-810b-c4580d6cf73b?currentLanguage=DE>

<https://yam-united.telekom.com/pages/group-privacy/apps/wiki/wiki/list/view/4ec95910-4af9-4bfe-86b5-d6e18e4a2140>

Transfer Impact Assessment (TIA) other legal entities

The TIA is a privacy related document. For this fact is has to be clared with the responsible DPO of the legal entity.

Pre Approved Services

Pre-approved services is on the roadmap of the Google Cloud LZ team, but **not avaiable yet**.

On the following link you will find a list of google services which are already approved: <https://cloudsecurity.telekom.de/de/magenta-cloud-community/service-status/>

The google landingzone team and the TSEC are currently working on the pre approval for google services. To give you an overview on which services we are working on see the following list:

Virtual Private Cloud	Big Query	DataProc	Compute Engine
Kubernetes Engine (GKE)	Cloud Data Loss Prevention	Datastore	Cloud Storage
Cloud KMS	App Engine	Cloud Composer	Pub/Sub
LoadBalancer	Cloud Run	Cloud Scheduler	Dataflow
Cloud Functions	Cloud Monitoring	Cloud Logging	Cloud SQL
Reverse Proxy	Cloud Tasks		

Not pre approved services can be used, but takes much more efforts in the PSA process.